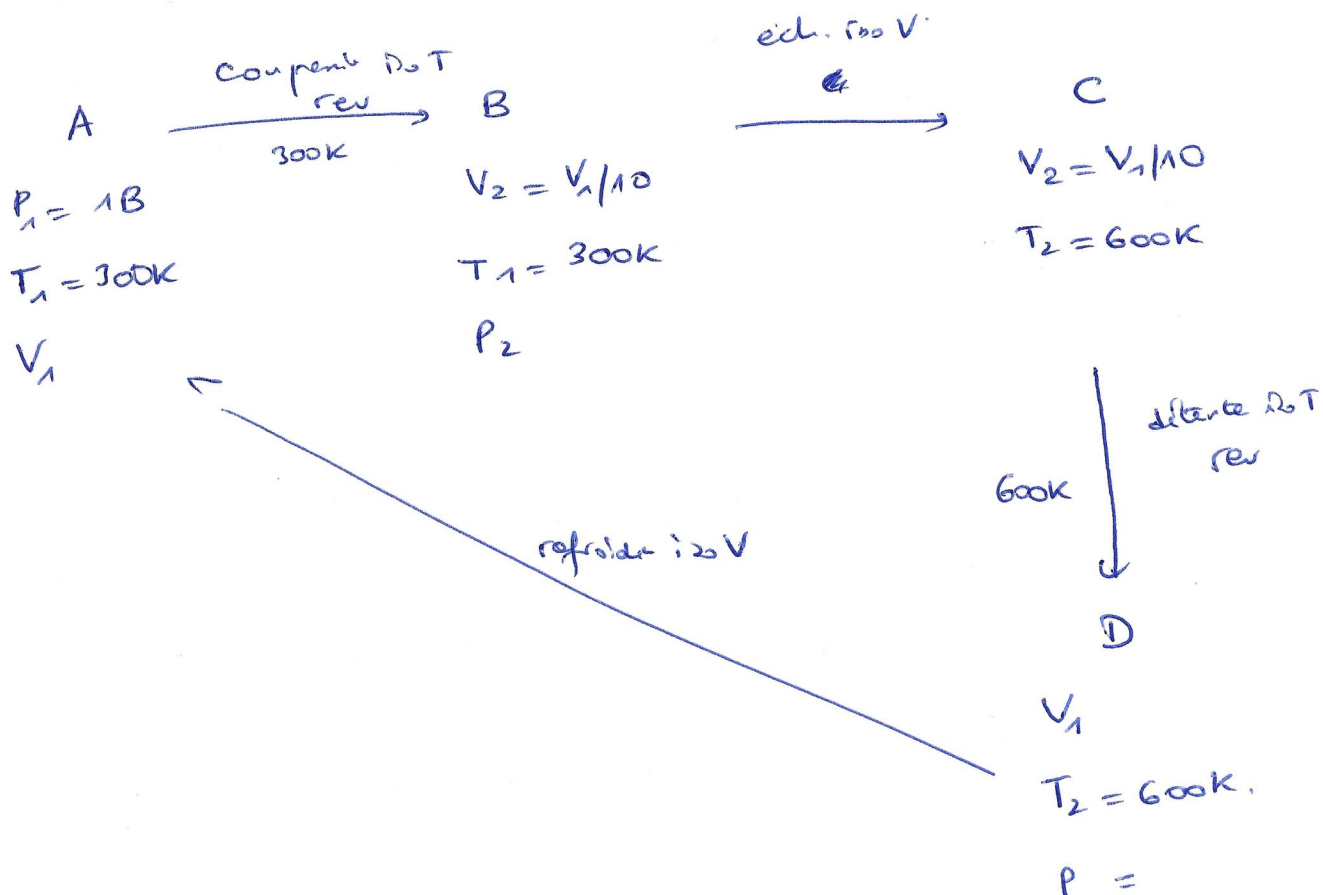


Machine de Stirling.



① état A:

$$P_1 V_1 = n R T_1$$

⇒

$$V_1 = \frac{n R T_1}{P_1}$$

$$\text{AN } V_1 = \frac{4 \cdot 10^{-2} \cdot 8,31 \cdot 300}{10^5} = 9,97 \cdot 10^{-4} \text{ m}^3 = 10 \cdot 10^{-3} \text{ m}^3 = \underline{1,0 \text{ L}}$$

état B

$$V_2 = 1,0 \cdot 10^{-1} \text{ L} = \underline{1,0 \cdot 10^{-4} \text{ m}^3}$$

$$P_2 V_2 = n R T_1$$

$$P_B = \frac{n R T_1}{V_2}$$

$$\text{AN: } P_B = \frac{4 \cdot 10^{-2} \cdot 8,31 \cdot 300}{10^{-4}} = 1,00 \cdot 10^6 \text{ Pa} = \underline{10 \text{ bar}}$$

etor C:

$$P_C V_2 = n R T_2$$

$$P_C = \frac{n R T_2}{V_2}$$

$$\text{AN: } P_C = \frac{4 \cdot 10^{-2} \cdot 8,31 \cdot 600}{10^{-4}}$$

$$= \underline{20,8}$$

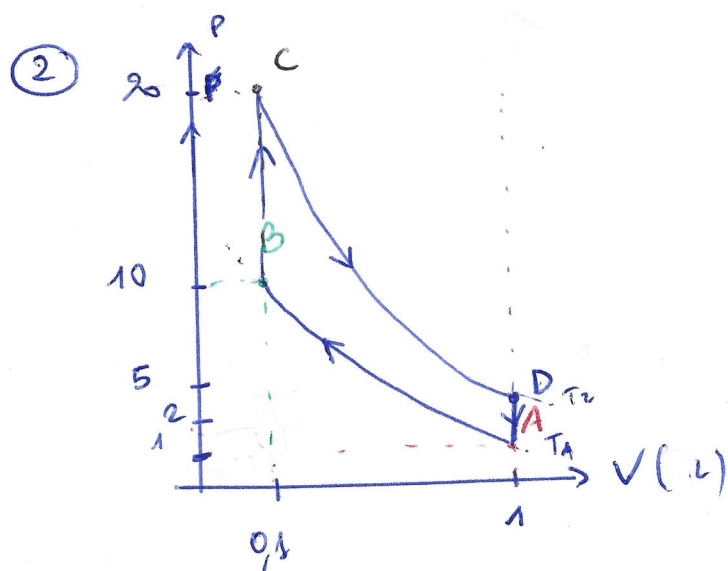
etor D:

$$P_D V_1 = n R T_2$$

$$P_D = \frac{n R T_2}{V_1}$$

$$P_D = \frac{8,31 \cdot 0,04 \cdot 600}{10^{-3}}$$

$$= \underline{2,03}$$



motor:

$$W = -\int P dV$$

$$= -A$$

$$W < 0$$

③ AB:

isot rev.

$$\Delta U = C_V \Delta T = 0 = W + Q$$

$$W_{AB} = -\int P dV = -nRT \int_{V_1}^{V_2} \frac{dV}{V} = -nRT \ln \frac{V_2}{V_1} = \underline{nRT \ln \frac{V_1}{V_2}}$$

$$Q_{AB} = -W_{AB} = \underline{nRT \ln \frac{V_2}{V_1}}$$

AN:

$$W_{AB} = 8,31 \cdot 0,04 \cdot 300 \ln \frac{1}{10} = \underline{+2,3 \cdot 10^2 \text{ J}}$$

$$Q_{AB} = \underline{-2,3 \cdot 10^2 \text{ J}}$$

BC:

$$\underline{W_{BC} = 0}$$

$$Q_{BC} = \Delta U = C_V \Delta T = C_V (T_2 - T_1).$$

$$Q_{BC} = \frac{nR}{\gamma - 1} (T_2 - T_1)$$

$$\begin{aligned} \text{AN: } Q_{BC} &= \frac{0,04 \cdot 8,31 (300)}{0,4} \\ &= \underline{2,5 \cdot 10^2 \text{ J}}. \end{aligned}$$

CD:

$$\Delta U = C_V \Delta T = 0 = W + Q$$

$$W_{CD} = -nRT \ln \frac{V_1}{V_2}$$

$$Q_{CD} = +nRT \ln \frac{V_1}{V_2}$$

$$\begin{aligned} \text{AN: } W_{CD} &= -8,31 \cdot 0,04 \cdot 600 \cdot \ln 10 \\ &= \underline{-4,6 \cdot 10^2 \text{ J}}. \end{aligned}$$

$$Q_{CD} = \underline{+4,6 \cdot 10^2 \text{ J}}.$$

DA: $W_{DA} = 0.$

$$Q_{DA} = \Delta U = C_V \Delta T = \frac{nR}{\gamma - 1} (T_1 - T_2)$$

$$\text{AN: } Q_{DA} = -Q_{BC} = \underline{-2,5 \cdot 10^2 \text{ J}}$$

	A	B	C	D
P	1,0B	10B	20B	2,0B
T	300K	300K	600	600
V	1,0 L	0,10 L	0,10 L	1,0 L

Source en contact transf.	S_1	S_2	S_2	S_1
	AB	BC	CD	DA
Q	$-2,3 \cdot 10^2$	$2,5 \cdot 10^2$	$4,6 \cdot 10^2$	$-2,5 \cdot 10^2$
W	$2,3 \cdot 10^2$	0	$-4,6 \cdot 10^2$	0

④ $W = -2,3 \cdot 10^2 J$

$Q_c = Q_{S_2} = 7,1 \cdot 10^2 J > 0$

$Q_F = Q_{S_1} = Q_{AB} + Q_{DA} = -4,8 \cdot 10^2 J < 0$

⑤ $\eta = \frac{-W}{Q_c}$

$= \frac{+2,3 \cdot 10^2}{7,1 \cdot 10^2} = 0,32$

⑥ $\Delta S = 0 = S_e + S_c \text{ (cycle)}$ $S_e = \frac{Q_c}{T_2} + \frac{Q_F}{T_1}$

$S_c = -\frac{Q_c}{T_2} - \frac{Q_F}{T_1}$

(thermostats)

AN: $S_c = -\frac{7,1 \cdot 10^2}{600} + \frac{4,8 \cdot 10^2}{300} = 0,42 J/K$

irréversibilité due aux inhomogénéités de T (diffusivité thermique).

⑦ On récupère Q_{DA} : l'énergie thermique est stockée ds le régénérateur.

donc $Q_c = Q_{CD} = 4,6 \cdot 10^2 \text{ J}$.

$$Q_{DA} = -Q_{BC}$$

On économise Q_{BC} .

⑧ $\eta' = \frac{-W}{Q_{CD}} = \frac{2,3 \cdot 10^2}{4,6 \cdot 10^2} = \underline{0,5} > 0,32.$

On a amélioré le rendement global de la machine!

⑨ $\eta_{\text{Carnot}} = 1 - \frac{T_1}{T_F}$

AN $\eta_{\text{Carnot}} = 1 - \frac{300}{600} = \underline{0,5}$

On a atteint le rendement Carnot

(rendement maximal pour 1 moteur diatherne).